

And the real “generating process” is extremely simple but it had nothing to do with a linear model! Some parts of it appear to be linear and we are fooled by extrapolating in a direct line.^{*}—

So not only can the past be misleading, but there are also many degrees of freedom in our interpretation of past events.

For the technical version of this idea, consider a series of dots on a page representing a number through time—the graph would resemble [Figure 1](#) showing the first thousand days in [Chapter 4](#). Let’s say your high school teacher asks you to extend the series of dots. With a linear model, that is, using a ruler, you can run only a straight line, a *single* straight line from the past to the future. The linear model is unique. There is one and only one straight line that can project from a series of points. But it can get trickier. If you do not limit yourself to a straight line, you find that there is a huge family of curves that can do the job of connecting the dots. If you project from the past in a linear way, you continue a trend. But possible future deviations from the course of the past are infinite.

This is what the philosopher Nelson Goodman called the riddle of induction: We project a straight line only because we have a linear model in our head—the fact that a number has risen for 1,000 days straight should make you more confident that it will rise in the future. But if you have a nonlinear model in your head, it might confirm that the number should decline on day 1,001.

Let’s say that you observe an emerald. It was green yesterday and the day before yesterday. It is green again today. Normally this would confirm the “green” property: we can assume that the emerald will be green tomorrow. But to Goodman, the emerald’s color history could equally confirm the “grue” property. What is this grue property? The emerald’s grue property is to be green until some specified date, say, December 31, 2006, and then blue thereafter.

The riddle of induction is another version of the narrative fallacy—you face an infinity of “stories” that explain what you have seen. The severity of Goodman’s riddle of induction is as follows: if there is no longer even a single unique way to “generalize” from what you see, to make an inference about the unknown, then how should you operate? The answer, clearly, will be that you should employ “common sense,” but your common sense may not be so well developed with respect to some Extremistan variables.

THAT GREAT ANTICIPATION MACHINE

The reader is entitled to wonder, So, NNT, why on earth do we plan? Some